

Indian Superstore Analytics Dashboard

End-to-End Power BI Business Intelligence Report

Project Scope: Data Preparation, Data Modeling [Star Schema],
Interactive Dashboards & Business Strategy

Submitted For: Academic Evaluation (Phases 1–5 Deliverables)

Tool Used: Power BI Desktop

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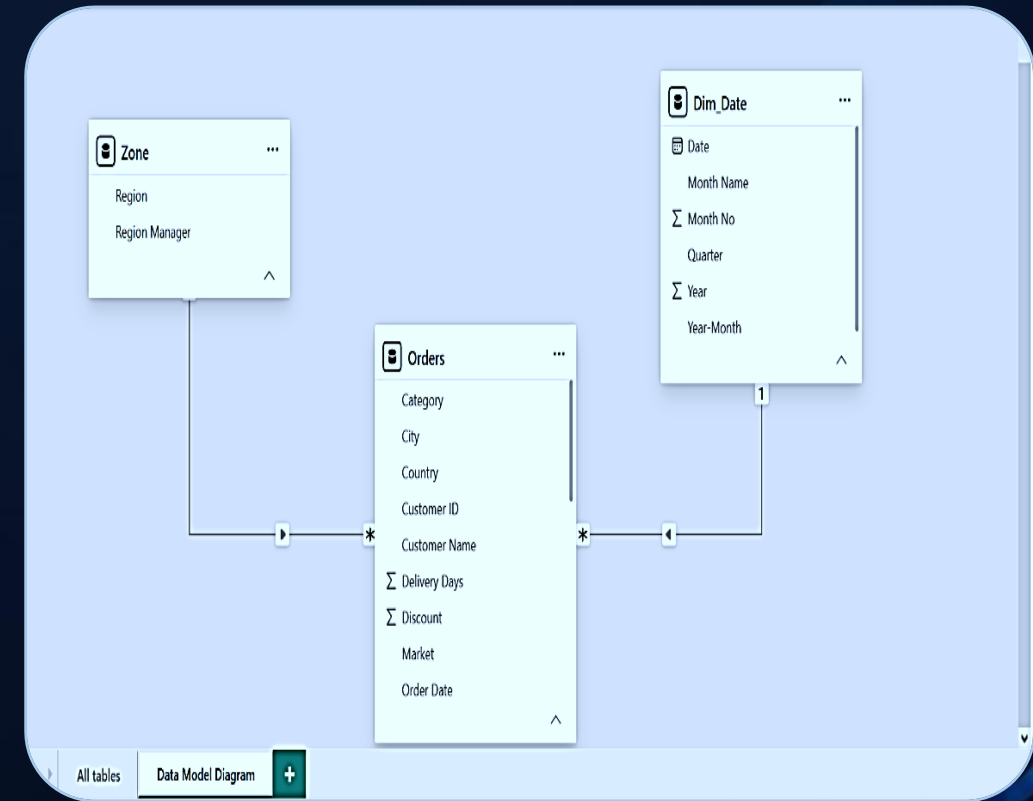


Project Scope & Architecture

- Phase 1: Power Query ETL
- Imported dataset, cleaned nulls/duplicates, fixed data types, and created calculated column Delivery Days.
- Phase 2: Data Modeling & DAX
- Built Star Schema connecting Orders fact table to Date, Customer, Product, and Region dimensions. Created key financial DAX measures.
- Phase 3: Multi-Page Dashboard
- Designed 4 core pages: Sales Overview, Customer, Product & Profitability, and Region/Manager View. Profile
- Phase 4 & 5: Advanced BI Features & Strategy
- Implemented Static RLS, Bookmarks, Custom Tooltips, Drill-Through, and executive business recommendations.

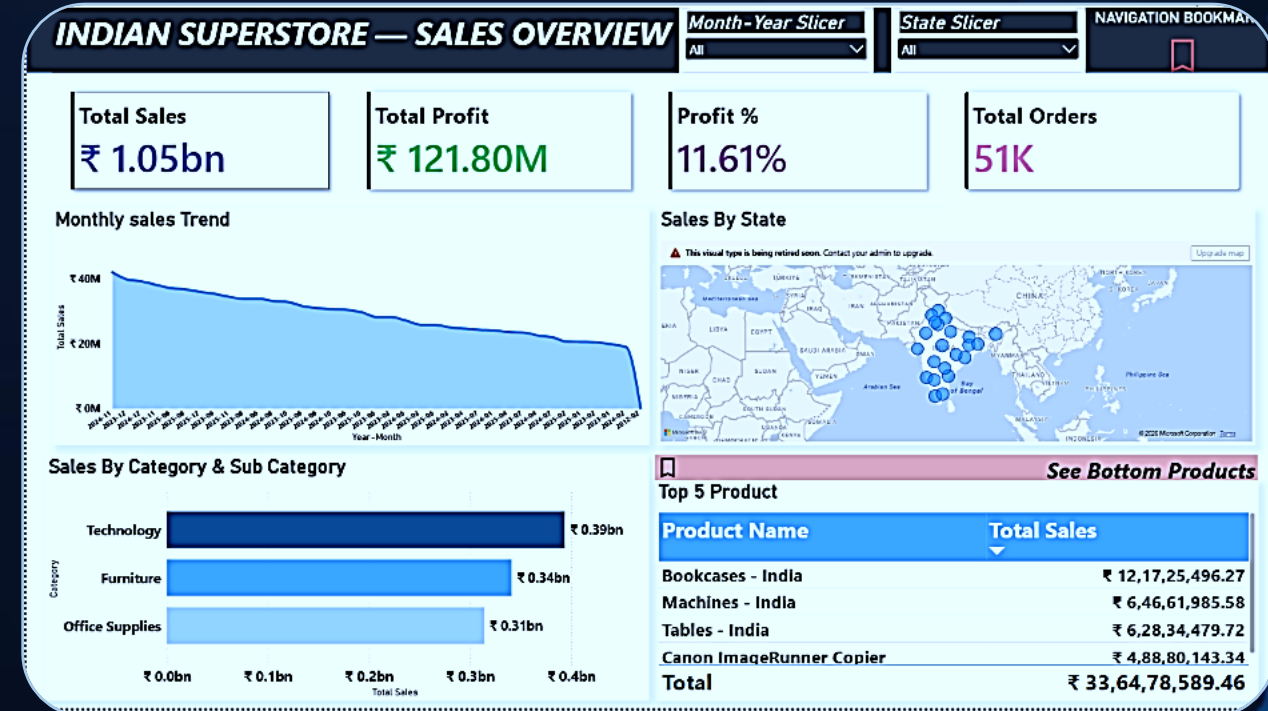
Phase 1 & 2: Data Cleaning & Star Schema Model

- Power Query ETL Steps (Phase 1):
- Imported dataset, set column data types (Date, Decimal, Text), removed duplicates/blanks.
- Added calculated column: $\text{Delivery Days} = [\text{Ship Date}] - [\text{Order Date}]$.
- Star Schema Architecture (Phase 2):
- Connected Orders fact table to Date, Customer, Product, and Region dimensions (1-to-Many single direction).
- Core DAX Measures Created:
- Total Sales, Total Profit, Profit %, Total Orders, Discount %, Delivery Days.



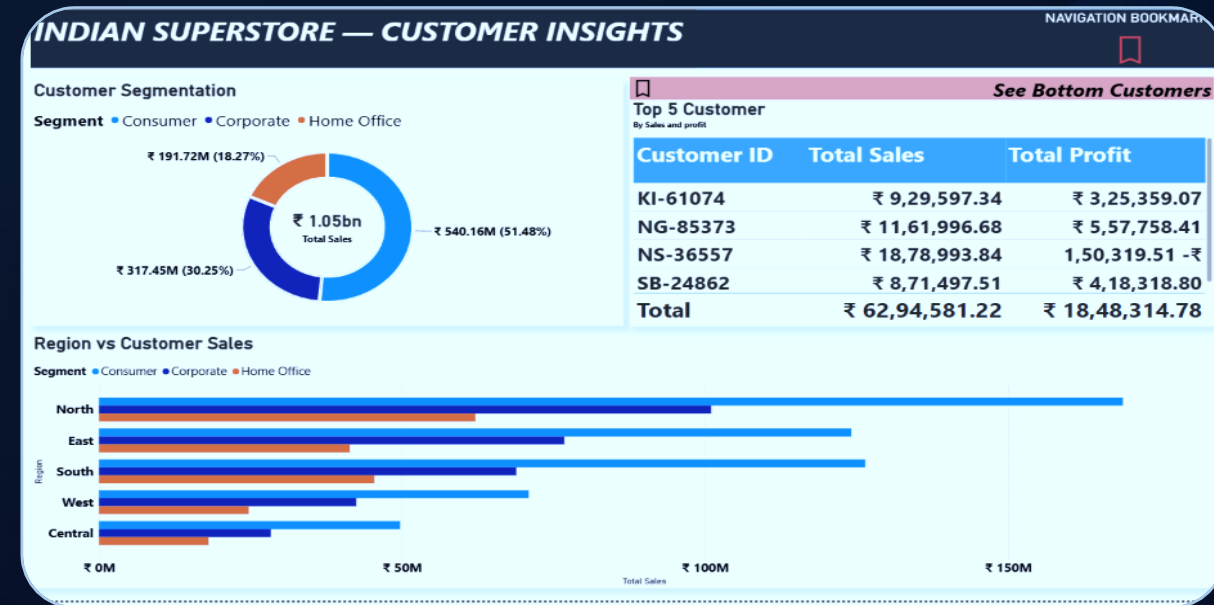
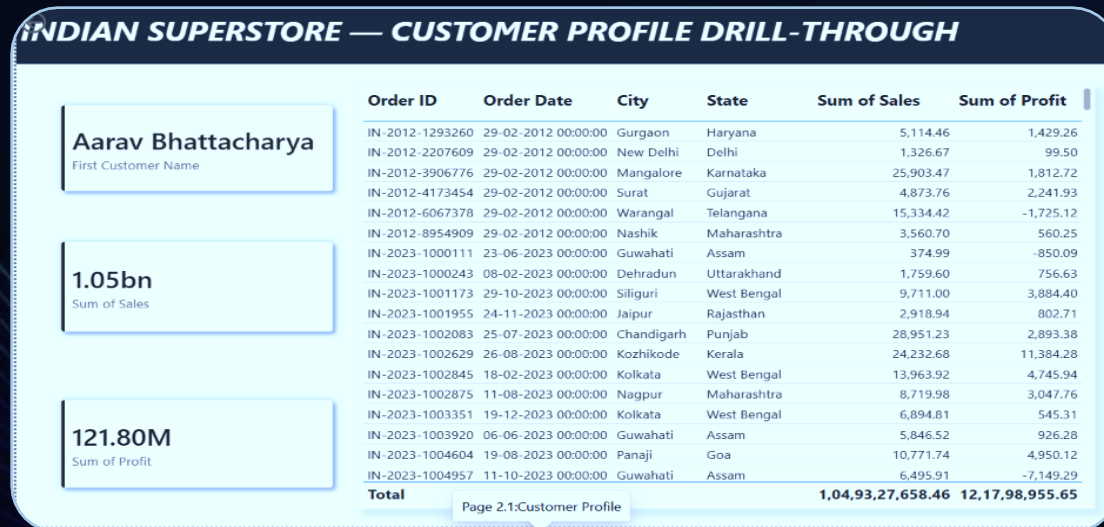
Phase 3: Page 1 – Sales Overview Insights

- Executive Revenue Metrics:
- Achieved ₹1.05bn Total Sales across 51K Total Orders.
- Delivered ₹121.80M Total Profit with an overall profit margin of 11.61%.
- Sales Dynamics & Distribution:
- Visualized revenue trends over time using the Monthly Sales Trend area chart.
- Tracked geographic sales density across regions using the Sales By State map visual.
- Category Performance:
- Categorized performance across Technology, Furniture, and Office Supplies via Category/Sub-Category bar charts.
- Listed top/bottom-performing items using the Top & bottom five Product breakdown table.



Phase 3: Page 2 – Customer Segment Insights

- INDIAN SUPERSTORE — CUSTOMER INSIGHTS):
- Customer Segmentation: Analyzed sales breakdown across Consumer, Corporate, and Home Office segments using the donut visual.
- High-Value Accounts: Identified top contributing accounts via Top/Bottom 5 Customer tables ranked by Total Sales and Total Profit.
- Regional Distribution: Evaluated regional customer purchasing behavior using Region vs Customer Sales bar breakdown.



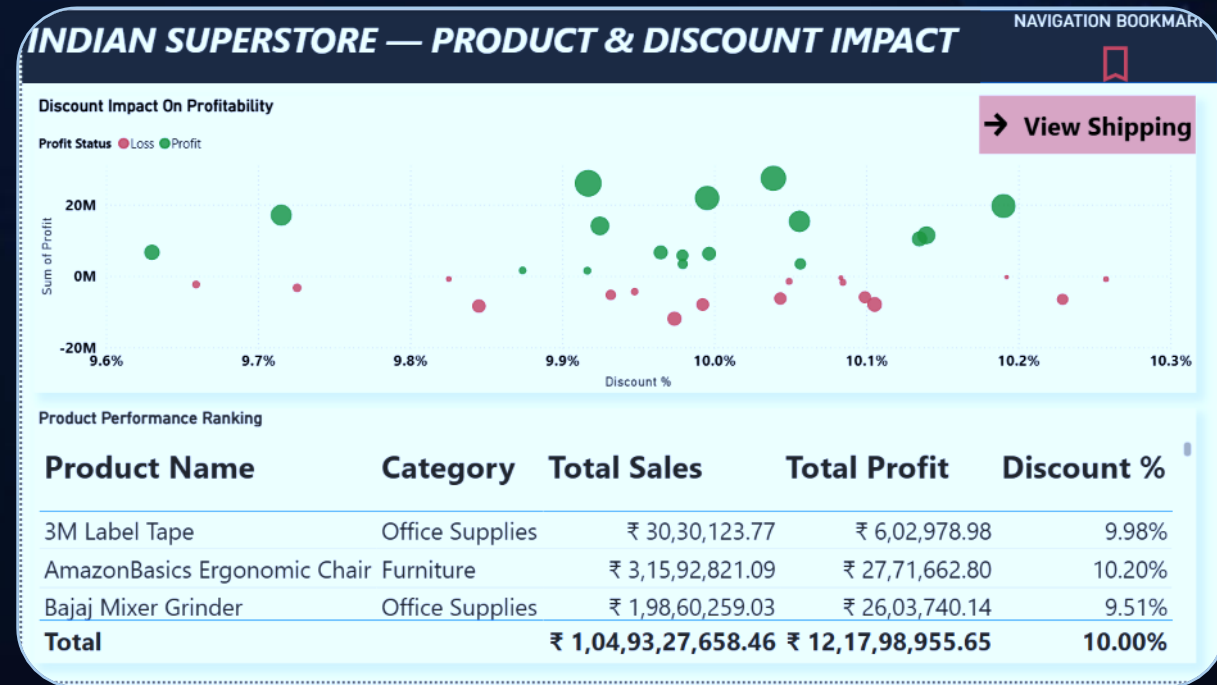
CUSTOMER PROFILE DRILL-THROUGH):

Interactive Drill-Through Feature: Configured Page 2.1: Customer Profile as a dedicated drill-through target.

Granular Transaction Details: Allows users to right-click any specific customer (e.g., Aarav Bhattacharya) on Page 2 to instantly view their detailed purchase history, order IDs, cities, states, sales, and profit metrics.

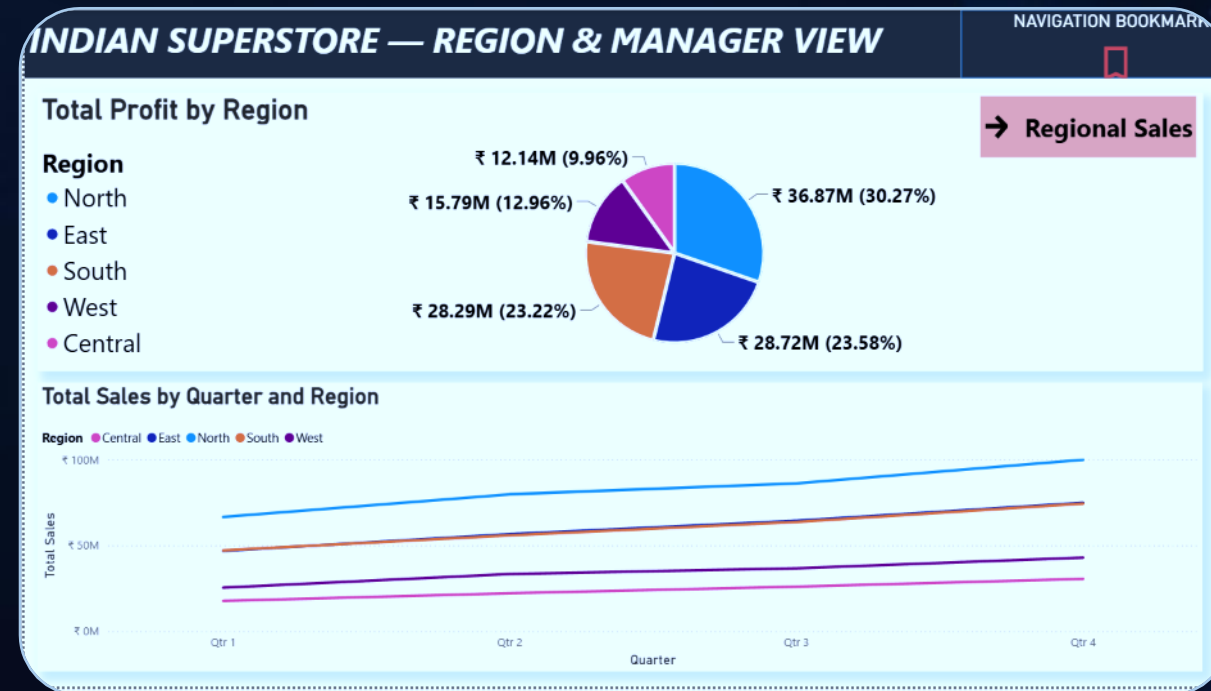
Phase 3: Page 3 – Product & Discount Profitability

- Product & Margin Analysis:
- Evaluated profitability across product categories, tracking overall sales revenue, net profit, and discount exposure.
- Detailed item performance in the Product Performance Ranking matrix (e.g., AmazonBasics, Ergonomic Chair, Bajaj Mixer Grinder).
- Discount Impact On Profitability:
- Analyzed scatter plot metrics revealing how higher discount percentages compress profit margins.
- View Switching via Bookmarks:
- Integrated interactive View Shipping and View Discount bookmark buttons directly on the page.
- Allows users to toggle seamlessly between the Average of Delivery Days by Ship Mode chart (Second Class, First Class, Standard Class, Same Day) and the Discount Impact On Profitability scatter view without changing pages



Phase 3: Page 4 – Region & Manager Overview

- Regional Performance Analysis:
- Analyzed sales and profit distribution across regions (North, East, South, West, Central) using donut charts and time-trend visuals.
- Identified top contributing regions (e.g., North and West leading in total sales and profit margins).
- Hierarchy Drill-Down Feature:
- Built hierarchy on the Total Sales by Region, State, City column chart.
- Enables dynamic drill-down capabilities from high-level Category down to granular Sub-Category level (and Region → State → City) for deep operational analysis.
- Regional View Switching:
- Integrated Regional Sales and Regional Profit bookmark buttons to switch views instantly for manager review.



Phase 4 & 5: Interactive Features & Business Strategy

- Consolidated Interactive Features Summary (Phase 4):
- Static Row-Level Security (RLS): Configured static regional manager roles (Central, East, South, North, West) restricting data access to authorized territories.
- Bookmarks & Header Navigation: Integrated header page-navigation buttons (Page 1 through Page 4) for seamless dashboard browsing, alongside view-switching toggles (View Shipping / View Discount and Regional Sales / Regional Profit).
- Customized Tooltips: Created dedicated tooltip pages (Custom Tooltip - P1, Custom Tooltip - P2) displaying pop-up previews on hover.
- Drill-Through Functionality: Configured right-click drill-through from Page 2 to Page 2.1: Customer Profile.
- Key Strategic Business Recommendations (Phase 5):
- Cap Discount Thresholds: Enforce a strict 15%–20% maximum discount limit on loss-making categories like Tables and Bookcases to protect net profit margins.
- Optimize Delivery Timelines: Streamline Standard Class fulfillment logistics to reduce the average shipping duration from 5–6 days.
- Replicate Regional Success: Expand high-margin marketing and inventory strategies from top regions (North/West) into Central and South territories.

Project Documentation & Author Profile

- Project Title: Indian Superstore Power BI Business Intelligence Analytics
- Developed By: **Mohd Sirajuddin Ahmed [Associate Data Scientist In Training]**
- Role: Data Analyst / BI Developer
- GitHub Repository: [Mohd Sirajuddin Ahmed GitHub](#)
- LinkedIn Profile: [Mohd Sirajuddin Ahmed LinkedIn](#)
- Project Portfolio Summary:
- Complete end-to-end Power BI report showcasing Power Query ETL, Star Schema modeling, dynamic DAX metrics, Row-Level Security (RLS), custom tooltips, and interactive drill-through features.
- Notice:
- "This presentation deck serves as visual documentation of the .pbix file for GitHub portfolio evaluation."